7A Perimeter and Area of Quadrilaterals and Composite Shapes

Standard Worksheet

Name: _____ Class: _____ Date: _____

PART A · FLUENCY & SKILL PRACTICE

Show complete step-by-step working in your workbook

Q1. **Convert each area measurement to the specified unit:**

(a) $7.2\ \text{cm}^2$ to mm^2 (b) $3.5\ \text{m}^2$ to cm^2 (c) $48\,000\ \text{m}^2$ to hectares (ha) (d) $1.6\ \text{km}^2$ to hectares (ha)

Q2. **Calculate the area of each parallelogram, rhombus, or kite:**

(a) Parallelogram: base $16\ \text{cm}$, perpendicular height $9\ \text{cm}$

(b) Parallelogram: base $12.5\ \text{m}$, perpendicular height $8\ \text{m}$

(c) Rhombus: diagonals $14\ \text{cm}$ and $18\ \text{cm}$

(d) Kite: diagonals $15\ \text{m}$ and $8\ \text{m}$

Q3. **Calculate the area of each trapezium with parallel sides a, b and perpendicular height h:**

(a) $a = 8\ \text{cm}, b = 14\ \text{cm}, h = 6\ \text{cm}$

(b) $a = 15\ \text{m}, b = 25\ \text{m}, h = 9\ \text{m}$

(c) $a = 7.5\ \text{cm}, b = 12.5\ \text{cm}, h = 8\ \text{cm}$

(d) $a = 19\ \text{m}, b = 11\ \text{m}, h = 12\ \text{m}$

Q4. **Find the unknown dimension in each shape:**

(a) Parallelogram: Area $= 135\ \text{cm}^2$, base $= 15\ \text{cm}$, find height h

(b) Kite: Area $= 84\ \text{cm}^2$, one diagonal $= 12\ \text{cm}$, find the other diagonal

(c) Trapezium: Area $= 90\ \text{m}^2, a = 7\ \text{m}, b = 11\ \text{m}$, find height h

(d) Trapezium: Area $= 140\ \text{cm}^2, h = 10\ \text{cm}, a = 12\ \text{cm}$, find parallel side b

PART B · PROBLEM-SOLVING & APPLICATIONS

Write clear mathematical statements with units

Q5. An L-shaped courtyard has outer dimensions $14\ \text{m}$ by $10\ \text{m}$, with a rectangular corner measuring $6\ \text{m}$ by $4\ \text{m}$ removed. Find (a) the perimeter of the courtyard and (b) the area of the courtyard.

Q6. A rectangular picture measuring $40\ \text{cm}$ by $30\ \text{cm}$ is surrounded by a timber frame of uniform width $5\ \text{cm}$ on all sides. Find the outer dimensions of the frame and the area of the timber frame itself.

Q7. A wheat farm in the Wimmera consists of a rectangular paddock of $600\ \text{m}$ by $400\ \text{m}$ joined to a triangular paddock of base $400\ \text{m}$ and perpendicular height $300\ \text{m}$. Find the total area of the farm in square metres and in hectares.

Q8. A kite-shaped stunt sail has diagonals of lengths $2.4\ \text{m}$ and $1.5\ \text{m}$. If ripstop nylon fabric costs $\$35$ per square metre, what is the cost of the fabric needed to make the sail?

PART C · REASONING & CHALLENGE

Justify your conclusions using algebraic or geometric reasoning

Q9. A parallelogram and a triangle have the exact same base length b and the exact same area A. How does the perpendicular height of the triangle compare to the perpendicular height of the parallelogram? Justify your answer algebraically.

Q10. Every staircase L-shape cut from the corner of a $W \times H$ rectangle has the same perimeter $2(W + H)$ as the original rectangle. Explain why removing a rectangular corner decreases the area but leaves the perimeter unchanged.